DiffORSINet: Salient Object Detection in Optical Remote Sensing Images via Conditional Diffusion Model

Yaoyao Hou and Ting Li

***Abstract*—Salient Object Detection in optical remote sensing images (ORSI-SOD) has received increasing attention in recent years. Although some progress has been made in existing methods, there are still challenges such as ambiguous and irregular boundaries of salient targets and complex backgrounds. The existing ORSI-SOD methods have difficulty in finely dividing the boundaries of salient targets and dealing with chaotic backgrounds. To solve these problems, we propose a new network based on the diffusion model, termed DiffORSINet, which describes the ORSI-SOD task as a conditional mask generation problem. By combining RGB images and the guidance of time steps, it can gradually and accurately locate and refine the segmentation of salient targets during the denoising process. Furthermore, we design a dedicated denoising network, which includes a Fourier frequency awareness module (FFAM) and a multilevel feature fusion module (MFFM), which significantly improves the refinement ability of the network. FFAM captures and fuses the frequency-domain features by combining the Fourier transform operation and the cross-attention (CA) mechanism, enhances the intensity of some signals, and thereby refines the image details. MFFM reduces the interference of chaotic backgrounds by coordinating and fusing multilevel features and suppressing irrelevant regions. Finally, the comparative experimental results on three widely used ORSI-SOD datasets show that the method proposed in this article is superior to other existing methods. Our code and results are available at https://github.com/hyy-qd/DiffORSINet/**

***Index Terms*—Diffusion model, Fourier frequency sensing, optical remote sensing images, salient object detection (SOD).**

I. Introduction

SALIENT Object Detection (SOD) aims to identify the most visually attractive objects or regions in a given visual scene [1], [2], [3], and plays an important role in various visual tasks, such as image/video segmentation [4], [5], [6], image quality assessment [7], object tracking [8], and medical imaging [9]. It also has a close collaborative relationship with image Captioning [10], providing spatial priors for captioning, such as remote sensing image captioning [11] and underwater image captioning [12]. SOD in Natural Scene Images (NSI-SOD) [13], [14], [15], [16] has achieved satisfactory development in recent years. As a result, more and more scientific researchers have applied SOD technology to optical remote sensing images. Saliency Object Detection in optical remote sensing images (ORSI-SOD) aims to detect attractive targets in optical remote sensing images, such as aircraft, vehicles, ships, and rivers. As an emerging research topic that has attracted much attention in recent years, it plays an important role in multiple fields such as military reconnaissance, environmental science, and security monitoring. Although the performance of the NSI-SOD algorithms has achieved relatively good results, natural scene images and optical remote sensing images are different in terms of resolution, object type, object proportion, and object scene. Therefore, when directly transferring the natural scene saliency object detection algorithm to the field of remote sensing images, satisfactory results often cannot be obtained. This article explores a specialized solution for salient object detection in optical remote sensing images, aiming to address the unique and challenging issues in optical remote sensing images.

Specifically, we adopt the diffusion model [17] as a novel solution for the Salient Object Detection in optical remote sensing images (ORSI-SOD) task. Due to its unique generation mechanism and modeling ability for complex data distribution, the diffusion model has more advantages over traditional methods in optical remote sensing image processing and has great potential in the detection of its inherent fuzzy boundaries and complex background interference. The conditional mask generation framework in the diffusion model represents the detection of salient object targets as a progressive enhancement process, and the mask in this framework iteratively generates noise through the input image. In the detection of salient targets in remote sensing images, traditional methods are prone to problems such as breakage or over-smoothing. However, the diffusion model can naturally handle the problem of blurred boundaries by gradually correcting the boundary area through multistep iterative denoising in the reverse diffusion process. Moreover, the diffusion model can directly learn the essential differences between the target and the background through its implicit feature decoupling and generative modeling capabilities. During the reverse denoising process, it can automatically separate the background texture interference from the core target features. Meanwhile, its conditional generation mechanism

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can actively suppress irrelevant background noise, significantly enhancing the robustness to complex scenes. Enable the model to maintain high-precision recognition in remote sensing images with complex backgrounds.

Notably, there are several challenges in processing optical remote sensing images. First, complex shooting conditions can cause the boundaries of prominent targets to become blurred and present irregular contours, making it difficult to accurately separate targets from the background. Therefore, it is very necessary to model the basic distribution of the data and deep explore the global and local information. However, there are certain limitations in directly applying the diffusion model to the ORSI task. Traditional diffusion models usually perform diffusion on images, but there is a significant difference between the image and the mask, which may lead to insufficient recognition ability of traditional diffusion models. Moreover, traditional diffusion models are typically based on CNN or Transformer. CNN is good at capturing local spatial details, but has limitations in learning global representations. In contrast, Transformer-based models can easily establish long-range dependencies to capture global relationships, but they are inferior to CNN in terms of local modeling.

To solve these problems, we propose a conditional guided denoising network based on the hybrid backbone network SMT [18], termed DiffORSINet. The DiffORSINet regards the ORSI-SOD task as a conditional mask generation task based on the diffusion model and diffuses the mask during the training process. In the denoising process, the image is used as a conditional cue to help improve the noise mask, and the hybrid backbone SMT combines the advantages of CNN and Transformer, having both the ability to fuse local details and global modeling. Meanwhile, in order to effectively remove noise interference and restore image details, we propose the Fourier Frequency Awareness Module (FFAM). **The FFAM module fully captures the frequency-domain features by using the Fast Fourier Transform (FFT) [19] operation, enhances the signal strength of some parts, highlights the edge and texture information of the image, and combines the cross-attention (CA) mechanism [20] to fuse the frequency-domain features, enhance local details, and determine the overall structure of the image, thereby achieving the effect of accurate target detection.**

Second, optical remote sensing images usually contain a large amount of noise interference and complex backgrounds. Since optical remote sensing images usually contain various natural landforms and artificial structures, these images typically show a chaotic background. In addition, factors such as lighting and weather can affect the imaging quality of optical remote sensing images. The targets in the images may be difficult to effectively distinguish from the background due to problems such as shadows and overexposure. Previous methods [21], [22], [23] used feature fusion methods to suppress background interference. However, in these methods, the interaction and complementarity among the features of different layers are ignored, often resulting in insufficient fusion of context information and a decrease in detection accuracy. **To solve this problem, we propose a Multilevel Feature Fusion Module (MFFM). This module coordinates multilevel features, seeks the correlations between global features and local features, and fuses high-level semantic information and low-level detail features step by step. This process can effectively filter out redundant background interference, suppress background noise, and highlight significant targets in the shallow representation.**

Our contributions are summarized as follows.

1) We specially develop an ORSI-SOD method based on the diffusion model. It adopts a hybrid backbone to extract features and combines the advantages of global and local modeling to understand the subtle differences between significant targets and the surrounding environment more comprehensively, enhance the processing ability of details, and improve the segmentation accuracy.
2) We propose a novel FFAM module to restore image details and perceive the boundary details of significant targets. It captures the frequency-domain features through the Fourier transform operator, reveals the hidden frequency information in the image, enhances the strength of some signals, and combines the CA mechanism to fuse different frequency features to achieve effective denoising, thereby improving the detection accuracy of the network.
3) We design a new MFFM module to mitigate the influence of a cluttered background. It achieves complementary advantages by coordinating multilevel features and gradually fusing the feature information of high and low layers, transmitting deep significant cues to shallow information units, suppressing irrelevant and chaotic background noise, and significantly improving the performance of target detection.
4) We conduct extensive experiments on three publicly available ORSI-SOD datasets to thoroughly evaluate our proposed model. The experimental results demonstrate the efficacy of each module within our framework, validating their contributions. Our model effectively addresses the inherent challenges in ORSI-SOD and achieves superior performance, outperforming 20 state-of-the-art (SOTA) methods in comprehensive comparisons.

II. Related Work

A. SOD for NSIs and ORSIs

In early studies, many methods were proposed to detect significant targets in NSI, and these methods mainly relied on manually designed features [24]. However, due to the subjectivity of the design and the limitations of its application, the performance based on manual design methods is often unsatisfactory. With the rapid development of deep learning technology, compared with traditional methods, it could avoid the complexity and subjectivity of manual feature extraction and automatically learn the optimal feature expression from the data. Feature representation methods based on neural networks greatly improved the performance of SOD [25], [26], [27]. However, among them, algorithms based on CNN faced challenges in effectively establishing remote relationships. With the emergence of Transformer, Transformer-based methods showed advantages in establishing remote relationships,

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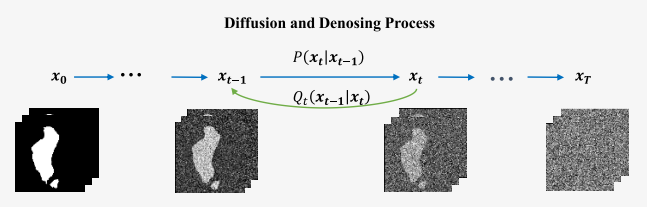


Fig. 1. Basic principle of the conditional diffusion model.

raising the performance of NSI-SOD to a new level. For instance, Zhuge et al. [16] proposed an integrity cognitive network to explore the learning of strong integrity features, determining whether the features of partial and overall objects had strong consistency through integrity channel enhancement and diverse feature aggregation. Liang et al. [28] proposed a multiscale attention and edge interactive refinement network to promote the close collaboration between attention and edge information, in order to improve the performance of SOD. Han et al. [29] designed a multiscale attention and edge interactive refinement network, which optimized the effect of SOD by enhancing the synergy between the attention mechanism and edge information.

Due to the inherent characteristic differences between NSI and ORSI, directly applying the NSI-SOD method to ORSI could not achieve good results. Therefore, researchers began to explore methods specifically targeting ORSI-SOD, raising the performance of ORSI-SOD to a new level. For instance, Zeng et al. [30] proposed an adaptive edge-aware semantic interaction network, which addressed issues such as complex backgrounds of salient targets, shadow interference, and complex edges through multiscale feature extraction and deep semantic interaction. Zhao et al. [21] captured global information and local details more effectively by combining CNN and Transformer, introduced a sparse attention mechanism to reduce noise interference and irrelevant information, and enhanced the model's ability to fuse different content information. Liu et al. [22] designed a hybrid heterogeneous encoder to extract heterogeneous features, combined with an adaptive global–local integration module and a salient-guided attention-enhancing decoder to locate and identify salient objects. Zhao et al. [23] proposed a recursive adaptive graph inference network, which reinforced the representation of boundary features in different scales and gradually interacted with global-context fusion information to capture significant targets in complex and irregular remote sensing images. Di et al. [31] introduced a multiscale and multidimensional weighted network to promote multiscale feature fusion and enhance detail and edge information, overcoming the challenges of large-scale variation, noise interference, and insufficient precise positioning caused by direct feature fusion. Sun et al. [32] proposed a local–global information awareness network based on Transformer, which enhanced fine-grained detail capture and broader context perception, guided precise edge generation, and improved salient target identification accuracy.

Although the aforementioned methods have made many improvements and achieved good progress in feature extraction, feature enhancement, and feature fusion, the challenges faced by optical remote sensing images are not completely solved, such as the irregular contours of salient targets and the interference of complex backgrounds. To address these issues, we propose a novel DiffORSINet, which includes a fast Fourier operation combined with the attention mechanism and a multiscale, multilevel feature step-by-step fusion module. This model demonstrates strong detection performance.

B. Diffusion Models

With the in-depth research on generative models, the architecture of diffusion models has become increasingly mature. Diffusion models that perform well in various computer vision tasks attracted people's attention and achieved breakthrough results in multiple fields, such as text [33], image [34], audio [35], and video [36]. The diffusion model [17] is a parameterized Markov chain. Its basic concept is to gradually add noise to make the data blurred and then gradually remove the noise to restore the original data or generate new data.

The diffusion model not only has outstanding performance and broad application prospects in image generation but also shows its powerful ability in object detection and image segmentation tasks. For instance, Wu et al. [37] proposed a new denoising diffusion model, DiffuMask. This method utilized the generative ability of the diffusion model and automatically acquired the accurate semantic mask of the synthesized image through a text-guided image synthesis process. Jiang et al. [38] proposed a camouflaged target segmentation method based on the bilateral diffusion model, transforming segmentation and depth estimation tasks into denoising tasks, and gradually restoring the target segmentation and depth information from the noisy image through the diffusion model. Chen et al. [39] regarded the detection of camouflaged targets as the task of generating conditional masks and utilized the iterative refinement process of the conditional diffusion model to enhance the model's ability to recognize the boundaries of fine objects, effectively suppressing overconfident erroneous predictions. Wu et al. [40] designed a new spectrum-space Transformer mechanism to integrate conditional semantic features and diffused noise features, thereby effectively narrowing the gap between anchoring conditions and semantic conditions. Wang et al. [41] proposed a new model called WaterCycleDiffusion, which achieves underwater image enhancement through vision-text fusion technology. It introduces differentiable underwater color differences as key enhancement cues and uses a discrete wavelet transform to decompose the image into low-frequency colors and high-frequency details. It effectively overcomes the shortcomings of scarce training data and reliance on manual selection of reference images. Xu et al. [42] proposed a comb concatenation diffusion model for super-resolution reconstruction of hyperspectral images. This method integrates conditional knowledge and noise in the feature space through the comb connection strategy, thereby enhancing the guiding ability of conditional knowledge for the denoising process and ingeniously handling the high dimension and complexity of hyperspectral images. Yan et al. [43] proposed an image enhancement model that utilizes the generative ability

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of diffusion models and frequency prior guidance. By applying frequency domain constraints to learn compact frequency priors, the iterative denoising process of the diffusion model is guided, and the Transformer blocks based on wavelets are used to enhance the network's generative ability, significantly accelerating reasoning and generating more accurate image content.

Although diffusion models have been applied in other fields, directly applying these models to the field of remote sensing cannot achieve the best results at present. Therefore, we propose a new conditional diffusion model, DiffORSINet. It combines a specially designed denoising network and an elaborately designed feature integration module, achieving more accurate detection of salient targets in optical remote sensing images.

III. Proposed Method

A. Network Overview

In this chapter, we will describe the specific process and details of DiffORSINet. The main idea of this method is shown in Fig. 1. DiffORSINet consists of two parts: the forward diffusion process and the reverse denoising process. During the forward diffusion process, noise is gradually added to the mask data through multiple iterations until the transformed data approaches pure noise for example, $P(x_t|x_0)$. Then, the reverse denoising process gradually removes the added noise through the denoising network for example, $Q(x_{t-1}|x_t)$ to obtain the denoising mask x_{t-1}. After the T-step denoising, the final saliency map is generated.

B. Diffusion Model

1) Forward Diffusion Process: Referring to the relevant thermodynamic formulas [44], this diffusion operates in the mode of a time-indexed Markov chain. Gradually add noise to the data within T time steps. When the time step is t, the noise mask x_t can be expressed by the following formula:

$$P(x_t|x_{t-1}) = \mathcal{N}\left(x_t;\ \sqrt{\alpha_t}x_{t-1},\ \sqrt{1-\alpha_t}\mathcal{I}\right) \tag{1}$$

where $\mathcal{N}$ represents the Gaussian distribution which specifies the mean and variance, $\mathcal{I}$ is the identity matrix, x_t and x_{t-1} represents the noise mask generated by adding the noise of time steps t and $t-1$. α_t is the noise scheduling parameter at the time step t, which controls the intensity of noise addition at each step. Specifically, when the original input x_0 is given, the marginal distribution of x_t at the time step t can be described as

$$\begin{cases} P(x_t|x_0) = \mathcal{N}\left(x_t;\ \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathcal{I}\right) \\ \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i. \end{cases} \tag{2}$$

2) Reverse Denoising Process: The reverse denoising process begins with a Gaussian distribution defined as $P(x_T) = \mathcal{N}(x_T; 0; \mathcal{I})$. Here, the denoising network $f_N(\cdot)$ systematically predicts and eliminates noises, generating accurate and fine segmentation masks. In each step of this reverse operation, guided by the image conditions, the noise mask x_t is denoised at each time step t to generate the previous noise mask x_{t-1}. Given the input image as a condition, the probability distribution of x_{t-1} can be expressed as follows:

$$\begin{cases} Q(x_{t-1}|x_t) = \mathcal{N}\left(x_{t-1}; \mu_N(x_t, t), \sigma_t^2 \mathcal{I}\right) \\ \sigma_t^2 = \dfrac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}(1-\alpha_t) \end{cases} \tag{3}$$

where σ_t^2 represents the variance of the conditional distribution $Q(x_{t-1}|x_t)$, and $\mu_N(x_t, t)$ represents the mean of the conditional distribution $Q(x_{t-1}|x_t)$. $\mu_N(x_t, t)$ can be expressed as

$$\mu_N(x_t, t) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)}{1-\bar{\alpha}_t} f_N(x_t, I, t) \tag{4}$$

where $f_N(\cdot)$ is the denoising network. The inputs of $f_N(\cdot)$ include the noisy mask x_t, input image I, and timestep t.

C. Denoising Network

The denoising network is designed to generate denoising mask predictions x_0' and intermediate masks x_{t-1} using a diffused method. This network is divided into two parts: the encoder and the decoder. The detailed structure is shown in Fig. 2. The encoder takes the noise mask x_t as input and extracts multilevel feature representations under the guidance of the input image and the time step t. Based on this, the decoder restores the detailed information with the help of the noise mask x_t, gradually predicts and reconstructs the denoised mask, and finally outputs the denoised mask x_0'.

1) Encoder: Due to the unique challenges that ORSI-SOD faces, such as the uncertainty of target scale and quantity, blurred boundaries, chaotic background, and the absence of significant objects, to solve the above difficulties, better capture the key detailed information of significant objects, and distinguish the subtle differences between them and the background, the feature extractor must have both global and local modeling capabilities simultaneously. Although the CNN commonly used in traditional diffusion models has advantages in local modeling that are conducive to refining details, due to the lack of a global perspective, it is vulnerable to interference from background noise, resulting in poor segmentation effects. Compared with CNN, the advantages of Transformer in global modeling make up for the deficiencies of CNN. However, it has a large number of parameters and a relatively high computational cost, which poses a challenge to the efficiency of the diffusion model. Therefore, our encoder selects a hybrid backbone SMT that combines the advantages of CNN and Transformer [13], which has both efficiency and the ability to integrate local and global modeling. In our framework, the encoder makes full use of the guiding role of image conditions, combining image conditions with noise masks to obtain a more powerful and richer feature representation.

We introduce the time step t to prompt the network to perform denoising at different time steps, and the noise level and recovery difficulty vary with a time step t. In order to pursue better results and a more suitable structure, we modify SMT. In the first layer of SMT, the Overlapping Patch Embedding (OPE) is designed to replace the traditional Patch Embedding (PE). By adaptively accepting noise data, I and x_t are mapped to the same channel dimension for basic addition operations

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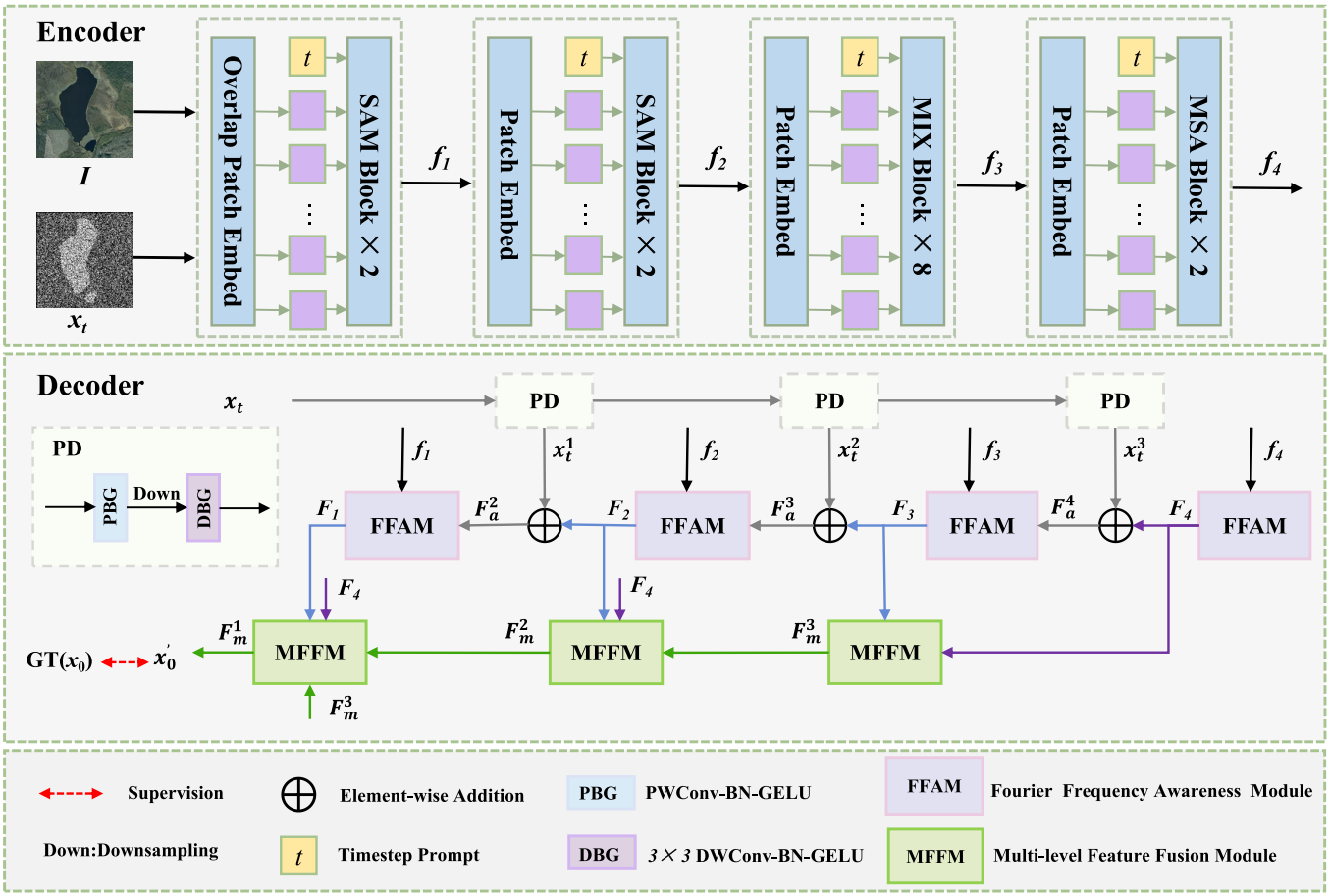


Fig. 2. Structure diagram of the denoising network of the DiffORSINet model. It consists of two parts: the encoder and the decoder. The encoder is responsible for extracting the local features and overall semantic information of the image. The decoder enhances the ability to express details by integrating and refining features from different levels and scales.

without destroying the original Transformer structure and pre-training parameters, and then the traditional PE operations continue. In the third stage, the combination of the original Scale-Aware Modulation (SAM) and Multihead Self-Attention (MSA), known as the MIX block, promotes effective interaction between local dependencies and global dependencies. In the fourth stage, MSA is utilized to obtain rich global semantic information. This process can be described as

$$f_i = \begin{cases} \mathrm{SAM}\left(\mathrm{Cat}\left(t, \mathrm{OPE}\left(I, x_t\right)\right)\right), & i = 1 \\ \mathrm{SAM}\left(\mathrm{Cat}\left(t, \mathrm{PE}\left(t, f_{t-1}\right)\right)\right), & i = 2 \\ \mathrm{MIX}\left(\mathrm{Cat}\left(t, \mathrm{PE}\left(t, f_{t-1}\right)\right)\right), & i = 3 \\ \mathrm{MSA}\left(\mathrm{Cat}\left(t, \mathrm{PE}\left(t, f_{t-1}\right)\right)\right), & i = 4 \end{cases} \tag{5}$$

where Cat(·) represents concatenation operation. OPE(·) refers to the OPE module, PE(·) denotes the PE module, SAM(·) indicates the SAM module, MIX(·) represents the MIX module, and MSA(·) refers to the MSA module, shown in Fig. 2.

D. Decoder

The decoder consists of the FFAM and the MFFM. In order to improve the refinement ability of the noise mask, we integrate the noise mask into the decoding process to help the network predict and remove noise more accurately. Among them, we use PBG (i.e., the combination of point-wise convolution layer, batch normalization layer and GELU activation function) and DBG (i.e., the combination of depth-wise convolution layer, batch normalization layer, and GELU activation function) to adjust the size and channels of the noise mask $x_t^i (i = 1, 2, 3)$.

This process can be described as follows:

$$\begin{cases} x_t^i = \mathrm{DBG}\left(\mathrm{PBG}\left(x_t\right)\right), & i = 1 \\ x_t^i = \mathrm{DBG}\left(\mathrm{PBG}\left(x_t^{i-1}\right)\right), & i = 2, 3 \end{cases} \tag{6}$$

where DBG(·) represents the DBG combination, and PBG(·) indicates the PBG combination.

1) Fourier Frequency Awareness Module: In response to the specific challenges of ORSI-SOD, such as variable target scales, complex structures, blurred boundaries caused by noise interference, and complex background problems, as shown in Fig. 3, we design the FFAM. This module aims to decouple features from the frequency dimension, group, enhance, and fuse them. The restoration of image details through selective enhancement enhances the network's ability to represent the irregular boundary details of complex scenes, providing accurate feature support for the subsequent decoder process.

Sum the input noise mask x_t^i with the corresponding FFAM layer feature F_{i+1} element by element to obtain feature

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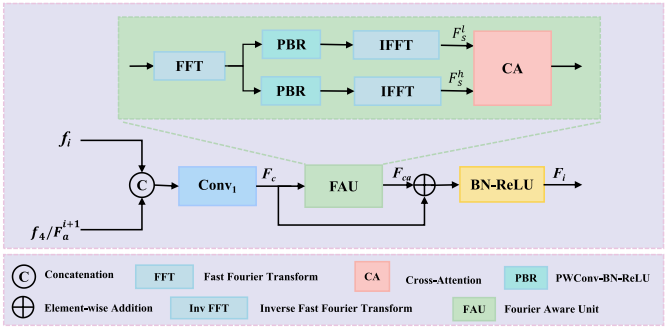


Fig. 3. Illustration of the FFAM.

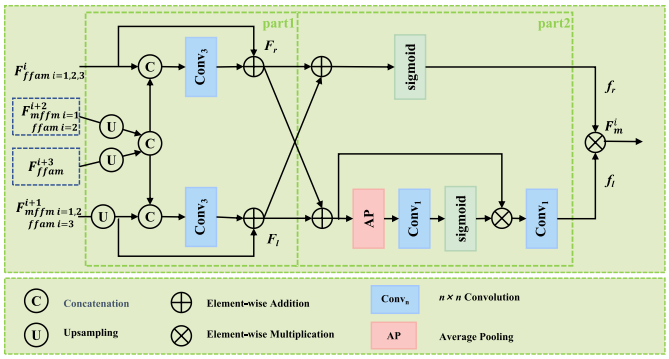


Fig. 4. Illustration of the MFFM.

F_a^{i+1}. FFAM generates F_i by fusing the feature information $f_i (i = 1, 2, 3)$ and F_a^{i+1} from the corresponding layers of the encoder. F_i is iterated as the input of the next-level FFAM. By means of this layer-by-layer iterative approach, FFAM can gradually enhance the expressive ability of features while retaining multiscale context information, thereby achieving more efficient feature integration. The entire process can be described as follows:

$$\begin{cases} F_a^{i+1} = x_t^i \oplus F_{i+1}, & i = 1, 2, 3 \\ F_i = \mathrm{FFAM}\left(f_i, f_i\right), & i = 4 \\ F_i = \mathrm{FFAM}\left(f_i, F_a^{i+1}\right), & i = 1, 2, 3 \end{cases} \tag{7}$$

where FFAM(·) represents the FFAM. ⊕ represents the sum of elements with each other.

The specific implementation method is shown in Fig. 3. When $i = 4$, two f_4 are cascaded. And when $i = 1, 2, 3$, the two inputs f_i and F_a^{i+1} of FFAM are cascaded to obtain F_c. Then, the spliced feature tensors are passed through the Fourier Awareness Unit (FAU) to obtain F_{ca}. Among them, we use the 2-D FFT operation to convert the signal from the spatial domain to the frequency domain, and in combination with the PBR operation, the F_c is divided into the low-frequency feature F_s^l and the high-frequency feature F_s^h. Low-frequency features contain the general contour of the image, while high-frequency features contain the detailed information of the image. Then we use the Inverse IFFT to restore the frequency signal to the spatial domain. To fuse the features in the frequency domain and the variation information in the suppression domain, we consider using the CA mechanism, which is widely used in multimodal feature fusion: the low-frequency features are used as queries ($\boldsymbol{Q}$), and the high-frequency features are used as keys ($\boldsymbol{K}$) and values ($\boldsymbol{V}$) to obtain the similarity weighting between the low-frequency features and the high-frequency features. The fusion of low-frequency and high-frequency features enables the determination of the overall structure of the image based on low-frequency information and the restoration of image details using high-frequency information, thereby more accurately locating the target and retaining detailed information. Finally, the output F_i of FFAM is obtained through the residual structure and BN-ReLU. The output matrix of CA can be expressed as

$$f_{\mathrm{CA}}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{QK}}{T\sqrt{d}}\right)\boldsymbol{V} = \boldsymbol{SV} \tag{8}$$

where d denotes the scaling factor and $\boldsymbol{S} \in \mathbb{R}^{N \times N}$ represents the attention weight matrix. Then the above process can be described as

$$\begin{cases} F_c = \mathrm{Conv}_1\left(\mathrm{Cat}\left(f_i, f_i\right)\right), & i = 4 \\ F_c = \mathrm{Conv}_1\left(\mathrm{Cat}\left(f_i, F_x^{i+1}\right)\right), & i = 1, 2, 3 \\ F_{\mathrm{ca}} = \mathrm{CA}\left(\mathcal{F}^{-1}\left(\mathrm{PBR}\left(\mathcal{F}\left(F_c\right)\right)\right)\right) \\ F_i = \mathrm{ReLU}\left(\mathrm{BN}\left(F_{\mathrm{ca}} \oplus F_c\right)\right) \end{cases} \tag{9}$$

where $\mathrm{Conv}_1(\cdot)$ represents 1×1 convolution, $\mathcal{F}(\cdot)$ indicates FFT, $\mathcal{F}^{-1}(\cdot)$ denotes IFFT, CA(·) refers to the CA mechanism, and PBR(·) represents the combination of point-wise convolution, batch normalization, and ReLU functions. BN(·) indicates batch normalization, and ReLU(·) denotes the ReLU function, and the remaining unexplained symbols indicate meanings consistent with the above.

2) Multilevel Feature Fusion Module: The structure of MFFM is illustrated in Fig. 4. During the decoding process, it is necessary to continuously integrate high-level semantic information and low-level detailed features, meticulously find the correlations between global features and local features, and capture the detailed features of the target more accurately. We design an MFFM module to generate the corresponding F_m^i by fusing the feature $F_i (i = 1, 2, 3)$ from the corresponding FFAM layer, the F_m^i of all previous fusion layers, and the top-level feature F_4. This model coordinates multilevel features, mines important semantic information, conducts refined processing of features at each layer, effectively suppresses background noise, and refines the boundaries of salient targets. The above process can be described as

$$F_m^i = \begin{cases} \mathrm{MFFM}\left(F_i, F_{i+1}\right), & i = 3 \\ \mathrm{MFFM}\left(F_i, F_m^{i+1}, F_{i+2}\right), & i = 2 \\ \mathrm{MFFM}\left(F_i, F_m^{i+1}, F_{i+3}, F_m^{i+2}\right), & i = 1 \end{cases} \tag{10}$$

where MFFM(·) represents the MFFM.

The specific structure of MFFM is shown in Fig. 4, which contains part 1 and part 2 internally. In part 1, after inputting the corresponding features of each layer and upsampling to adjust the size, they are concatenated. The first and second features of the input are used as the left and right branches, and the features of other layers (if any) are concatenated to the left and right branches in sequence. Then, through a 3×3 convolution and a residual connection, the features F_l and F_r

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of the left and right branches are obtained. Then proceed to part 2. After adding the left branch feature F_l and the right branch feature F_r, the information fusion features are extracted through cascaded average pooling, a 1×1 convolutional layer, and Sigmoid activation. The added values are multiplied element by element for calibration. Subsequently, the 1×1 convolutional layer is used to retain important features and discard features with less information to obtain f_l. Meanwhile, the left branch feature F_l and the right branch feature F_r are added, respectively, through 1×1 convolutional layers and activated by Sigmoid to obtain f_r. Ultimately, f_l is multiplied by f_r element by element to obtain F_m^i. The above process can be described as

$$\begin{cases} F_r = \mathrm{Conv}_3\left(F_i\right) \oplus F_i, & i = 3 \\ F_l = \mathrm{Conv}_3\left(F_{i+1}\right) \oplus F_{i+1}, & i = 3 \\ F_r = \mathrm{Conv}_3\left(\mathrm{Cat}\left(F_i, U\left(F_{i+2}\right)\right)\right) \oplus F_i, & i = 2 \\ F_l = \mathrm{Conv}_3\left(\mathrm{Cat}\left(U\left(F_m^{i+1}\right), U\left(F_{i+2}\right)\right)\right) \oplus U\left(F_m^{i+1}\right), & i = 2 \\ F_r = \mathrm{Conv}_3\left(\mathrm{Cat}\left(F_i, U\left(F_{i+3}\right), U\left(F_m^{i+2}\right)\right)\right) \oplus F_i, & i = 1 \\ F_l = \mathrm{Conv}_3\left(\mathrm{Cat}\left(U\left(F_m^{i+1}\right), U\left(F_{i+3}\right)\right.\right. \\ \qquad \left.\left. U\left(F_m^{i+2}\right)\right)\right) \oplus U\left(F_m^{i+1}\right), i = 1 \end{cases} \tag{11}$$

$$\begin{cases} f_l = \mathrm{Conv}_1\left(\mathrm{Sig}\left(\mathrm{Conv}_1\left(\mathrm{AP}\left(F_r \oplus F_l\right)\right)\right)\right) \otimes \left(F_r \oplus F_l\right) \\ f_r = \mathrm{Sigmoid}\left(\mathrm{Conv}_1\left(F_r\right)\right) \oplus \mathrm{Conv}_1\left(F_l\right) \\ F_m^i = f_l \otimes f_r, \quad i = 1, 2, 3 \end{cases} \tag{12}$$

where $\mathrm{Conv}_3(\cdot)$ denotes 3×3 convolution, $U(\cdot)$ represents upsampling, AP(·) refers to average pooling, and Sigmoid(·) indicates Sigmoid activation. The remaining unexplained symbols indicate meanings consistent with the above.

E. Loss Function

To enhance feature learning and obtain more accurate prediction graphs, we use mixed loss to train our model. This loss function consists of Binary Cross-Entropy (BCE) loss [45], Intersection over Union (IoU) loss [46], and F-measure (F-m) loss [47]. It pays attention to pixel-level details, image-level losses, and foreground and background differences simultaneously, and can comprehensively evaluate the differences between the predicted result map and the corresponding ground truth (GT). We present the total loss L of the following model:

$$L = L_{\mathrm{BCE}}\left(P_{t-1}, x_0\right) + L_{\mathrm{IoU}}\left(P_{t-1}, x_0\right) + L_{\text{F-loss}}\left(P_{t-1}, x_0\right) \tag{13}$$

where P_{t-1} is the prediction graph, x_0 is the truth graph, L_{BCE}, L_{IoU}, and $L_{\mathrm{F-loss}}$ are the BCE loss, IoU loss, and F-m loss, respectively.

IV. Experiments

A. Experimental Setup

1) Datasets: We conduct experiments on the ORSSD [48], EORSSD [49], and ORSI-4199 [50] datasets. The ORSSD dataset, as the first public dataset of ORSI-SOD, contains 800 images and the corresponding pixel-level GT. Among them, 600 images are used for training and 200 images are used for testing. The EORSSD dataset is an extended version of ORSSD, with more diverse and challenging data samples. It contains 2000 images and corresponding GT, among which 1400 images are used for training and 600 images for testing. The ORSI-4199 dataset is the largest dataset of ORSI-SOD, containing 4199 images and the corresponding GT. Among them, 2000 images are used for training and 2199 images are used for testing. We train DiffORSINet on the training set of each dataset and test it on the corresponding test set.

2) Experimental Details: We train our model using a single NVIDIA RTX 4090 GPU through PyTorch. During the training process, the size of the input image is adjusted to 384×384. We apply a series of data augmentation techniques, combining random flipping and rotation, and random cropping to the training image, and expand seven additional variations in the original training data. The size of the training set becomes eight times the original. The encoder network adopts the modified SMT model and is initialized with pre-training parameters. Furthermore, we use the Adam optimizer. The batch size is 8, the initial learning rate is set to 1e-4, and it decays to 1/10 every 40 epochs. A total of 100 epochs are trained. In addition, we set the number of time steps $T = 5$ for the sampling process.

3) Evaluation Indicators: We adopt some widely used indicators to comprehensively evaluate the performance of the model, including S-measure (S_α, $\alpha = 0.5$) [51], F-m ($F_\beta^{\max}, F_\beta^{\mathrm{mean}}, F_\beta^{\mathrm{adp}}, \beta^2 = 0.3$) [52], E-measure ($E_\xi^{\max}, E_\xi^{\mathrm{mean}}, E_\xi^{\mathrm{adp}}$) [53], and mean absolute error (MAE, $\mathcal{M}$) [54]. The Precision-Recall (PR) curve and F-m curve quantitatively evaluate the performance of our method and all comparison methods on three datasets.

B. Comparison With SOTAs

We compare DiffORSINet with 20 SOTA NSI-SOD and ORSI-SOD methods on the EORSSD and ORSSD datasets, including MCCNet [47], ACCoNet [55], ICON [16], AESINet [30], SeaNet [56], GeleNet [57], SFANet [58], PRNet [59], HFCNet [22], MEANet [60], ASNet [61], MCP-Net [62], ADSTNet [21], RAGRNet [23], WeightNet [31], SOLNet [63], BCARNet [64], and LGIPNet [32]. Diffusion models for image segmentation methods include FocusDiff [65] and CamoDiff [39]. For the ORSI-4199 dataset, some model results of this dataset are still being refined. Therefore, we choose to present a set of representative results. The saliency maps of the above-mentioned methods are generated by running public source codes or derived from authors' and public benchmark tests.

1) Quantitative Evaluation: We report in Table I, the quantitative comparison results of DiffORSINet and other comparison methods on the EORSSD and ORSSD datasets. Overall, our method demonstrates outstanding performance on both public datasets, outperforming all the compared methods. Specifically, on the ORSSD dataset, all six metrics of DiffORSINet (S_α, $\mathcal{M}$, $F_\beta^{\max}$, F_β^{mean}, F_β^{adp}, and E_ξ^{mean}) are optimal and dominant. For $\mathcal{M}$ and F_β^{adp}, the values of DiffORSINet are 0.08% and 1.36% higher than those of the second-place RAGRNet [23], respectively. Regarding S_α, $F_\beta^{\max}$, F_β^{mean} and

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TABLE I

QUANTITATIVE COMPARISONS WERE MADE ON THE EORSSD AND ORSSD DATASETS WITH THE SOTA NSI-SOD AND ORSI-SOD METHODS. ↓ INDICATES THAT THE LOWER, THE BETTER, WHILE ↑ IS THE OPPOSITE. WE MARK THE FIRST TWO RESULTS IN RED AND BLUE, RESPECTIVELY. F_β^m AND E_ξ^m DENOTE F_β^{MEAN} AND E_ξ^{MEAN}, RESPECTIVELY

Methods	Params↓ (M)	FLOPs↓ (G)	FPS↑ (GPU)	EORSSD $S_\alpha\uparrow$	EORSSD $\mathcal{M}\downarrow$	EORSSD $F_\beta^{max}\uparrow$	EORSSD $F_\beta^{m}\uparrow$	EORSSD $F_\beta^{adp}\uparrow$	EORSSD $E_\xi^{max}\uparrow$	EORSSD $E_\xi^{m}\uparrow$	EORSSD $E_\xi^{adp}\uparrow$	ORSSD $S_\alpha\uparrow$	ORSSD $\mathcal{M}\downarrow$	ORSSD $F_\beta^{max}\uparrow$	ORSSD $F_\beta^{m}\uparrow$	ORSSD $F_\beta^{adp}\uparrow$	ORSSD $E_\xi^{max}\uparrow$	ORSSD $E_\xi^{m}\uparrow$	ORSSD $E_\xi^{adp}\uparrow$
ORSI-SOD Method and NSI-SOD Method																			
$MCCNet_{22}$	67.7	117.0	102	0.9327	0.0066	0.8904	0.8604	0.8137	0.9755	0.9685	0.9538	0.9437	0.0087	0.9155	0.9054	0.8957	0.9800	0.9758	0.9735
$ACCoNet_{22}$	102.56	348.8	84	0.9290	0.0074	0.8837	0.8552	0.7969	0.9727	0.9653	0.9450	0.9437	0.0088	0.9149	0.8971	0.8806	0.9796	0.9754	0.9721
$ICON_{23}$	33.1	2.9	121	0.9185	0.0073	0.8622	0.8371	0.8065	0.9687	0.9619	0.9497	0.9256	0.0116	0.8939	0.8671	0.8444	0.9704	0.9637	0.9554
$AESINet_{23}$	41.1	55.0	-	0.9358	0.0079	0.8838	0.8524	0.7923	0.9751	0.9636	0.9462	0.9460	0.0086	0.9183	0.8986	0.8666	0.9828	0.9747	0.9707
$SeaNet_{23}$	2.8	1.7	354	0.9208	0.0073	0.8649	0.8519	0.8304	0.9710	0.9651	0.9602	0.9260	0.0105	0.8942	0.8772	0.8625	0.9767	0.9722	0.9670
$GeleNet_{23}$	25.5	11.7	110	0.9376	0.0064	0.8923	0.8781	0.8641	0.9828	0.9766	0.9750	0.9469	0.0079	0.9254	0.9128	0.9035	0.9860	0.9815	0.9786
$SFANet_{24}$	25.1	7.7	-	0.9349	0.0058	0.8833	0.8680	0.8492	0.9769	0.9726	0.9669	0.9453	0.0077	0.9192	0.9063	0.8984	0.9830	0.9789	0.9765
$PRNet_{24}$	20.8	8.5	-	0.9296	0.0053	0.8707	0.8510	0.8282	0.9809	0.9732	0.9625	0.9468	0.0068	0.9153	0.9013	0.8915	0.9866	0.9824	0.9779
$HFCNet_{24}$	149.0	160.7	35	0.9449	0.0049	0.8951	0.8646	0.8051	0.9842	0.9720	0.9552	0.9543	0.0070	0.9269	0.9049	0.8710	0.9884	0.9798	0.9748
$MEANet_{24}$	3.3	9.6	133	0.9282	0.0070	0.8766	0.8678	0.8557	0.9708	0.9658	0.9663	0.9340	0.0098	0.9033	0.8934	0.8860	0.9768	0.9730	0.9715
$ASNet_{24}$	65.4	20.1	-	0.9345	0.0055	0.8959	0.8770	0.8672	0.9783	0.9745	0.9748	0.9441	0.0081	0.9172	0.9072	0.8986	0.9803	0.9764	0.9795
$MCP\text{-}Net_{24}$	25.1	86.6	-	0.9373	0.0070	0.8868	0.8620	0.8077	0.9765	0.9657	0.9515	0.9433	0.0090	0.9135	0.8979	0.8726	0.9807	0.9736	0.9724
$ADSTNet_{24}$	62.1	27.7	-	0.9373	0.8868	0.8620	0.8077	0.9765	0.9657	0.9515	0.0070	0.9433	0.9135	0.8979	0.8726	0.9807	0.9736	0.9724	0.0090
$RAGRNe_{24}$	35.6	17.8	-	0.9361	0.0057	0.8852	0.8770	0.8644	0.9788	0.9737	0.9749	0.9507	0.0066	0.9242	0.9157	0.9104	0.9861	0.9818	0.9800
$WeightNet_{24}$	103.7	143.9	-	0.9427	0.0053	0.8955	0.8800	0.8437	0.9808	0.9760	0.9674	0.9536	0.0068	0.9281	0.9159	0.9043	0.9863	0.9826	0.9801
$SOLNet_{25}$	6.5	8.1	289	0.9171	0.0078	0.8609	0.8513	0.8392	0.9623	0.9577	0.9596	0.9284	0.0111	0.9012	0.8946	0.8925	0.9734	0.9688	0.9728
$BCARNet_{25}$	24.0	7.0	-	0.9360	0.0054	0.8871	0.8735	0.8740	0.9761	0.9727	0.9722	0.9466	0.0071	0.9196	0.9070	0.9073	0.9833	0.9798	0.9832
$LGIPNet_{25}$	52.1	47.0	-	0.9392	0.0053	0.8924	0.8770	0.8623	0.9806	0.9753	0.9719	0.9507	0.0066	0.9266	0.9129	0.9027	0.9876	0.9829	0.9809
Diffusion-based Methods																			
$FocusDiff_{24}$	166.2	3810.1	3	0.9184	0.0091	0.8536	0.8429	0.8405	0.9591	0.9507	0.9450	0.9352	0.0101	0.9195	0.8896	0.8605	0.9707	0.9550	0.9652
$CamoDiff_{24}$	71.8	60.3	8	0.9298	0.0075	0.8809	0.8610	0.8715	0.9685	0.9484	0.9673	0.9410	0.0096	0.9116	0.8913	0.8908	0.9779	0.9651	0.9706
Ours	17.6	14.1	19	0.9450	0.0038	0.8994	0.8912	0.8921	0.9848	0.9800	0.9564	0.9550	0.0058	0.9287	0.9218	0.9240	0.9881	0.9844	0.9681

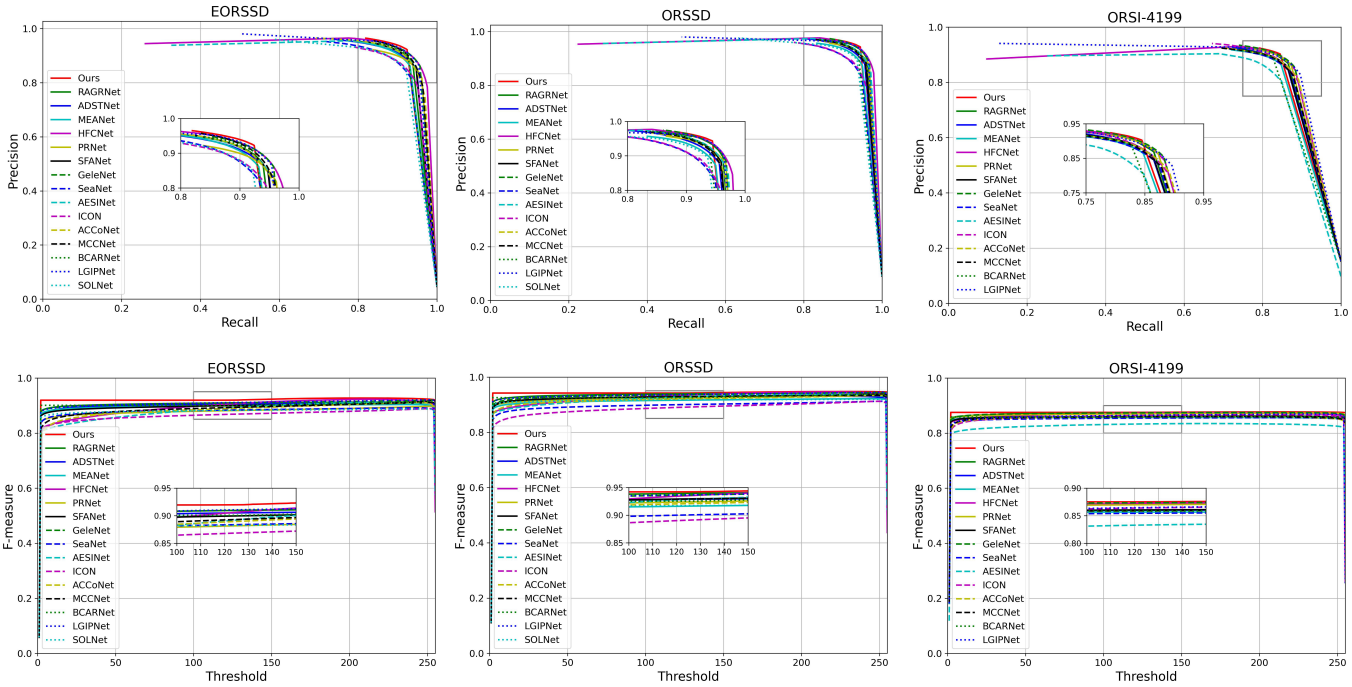


Fig. 5. P-R curves and F-m curves of different methods. Compared with other methods, our model achieved the highest accuracy and recall rate scores on three benchmark datasets.

E_ξ^{mean}, our method is 0.07%, 0.06%, 0.59%, and 0.15% higher than the second-ranked method. On the EORSSD dataset, our method outperforms all comparison methods in seven metrics (S_α, $F_\beta^{\max}$, F_β^{mean}, F_β^{adp}, $E_\xi^{\max}$, E_ξ^{mean}). Specifically, for the M metric, our method outperforms the second-ranked method by 0.11%. For the F-m metrics including $F_\beta^{\max}$, F_β^{mean} and F_β^{adp}, our method outperforms the second-ranked method by 0.35%, 1.12%, and 1.81%, respectively. When the E-measure metrics include $E_\xi^{\max}$ and E_ξ^{mean}, DiffORSINet is 0.06% and 0.34% higher than the second-place method,

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TABLE II

QUANTITATIVE COMPARISONS WERE MADE ON THE ORSI-4199 DATASETS WITH THE SOTA NSI-SOD AND ORSI-SOD METHODS. ↓ INDICATES THAT THE LOWER, THE BETTER, WHILE ↑ IS THE OPPOSITE. WE MARK THE FIRST TWO RESULTS IN RED AND BLUE, RESPECTIVELY

Methods	ORSI-4199							
	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{max}\uparrow$	$F_\beta^{m}\uparrow$	$F_\beta^{adp}\uparrow$	$E_\xi^{max}\uparrow$	$E_\xi^{m}\uparrow$	$E_\xi^{adp}\uparrow$
MCCNet	0.8746	0.0316	0.869	0.863	0.8592	0.9413	0.9348	0.9182
ACCoNet	0.8775	0.0314	0.8686	0.862	0.8581	0.9412	0.9342	0.9167
ICON	0.8752	0.0282	0.8763	0.8664	0.8531	0.9521	0.9438	0.9239
AESINet	0.8702	0.0325	0.8670	0.8573	0.8251	0.9403	0.9310	0.9184
SeaNet	0.8722	0.0308	0.8653	0.8591	0.8556	0.9426	0.9363	0.9197
GeleNet	0.8862	0.0264	0.8842	0.8785	0.8755	0.9544	0.9478	0.9265
SFANet	0.8761	0.0292	0.8710	0.8659	0.8647	0.9447	0.9385	0.9418
PRNet	0.8822	0.0254	0.8816	0.8750	0.8715	0.9539	0.9499	0.9519
HFCNet	0.8828	0.0281	0.8832	0.8720	0.8323	0.9525	0.9432	0.9246
MEANet	0.8707	0.0311	0.8662	0.8624	0.8638	0.9417	0.9338	0.9390
ADSTNet	0.8710	0.0318	0.8698	0.8653	0.8655	0.9433	0.9356	0.9212
RAGRNet	0.8811	0.0284	0.8811	0.8763	0.8758	0.9492	0.9428	0.9261
WeightNet	0.8912	0.0260	0.8873	0.8804	0.8756	0.9540	0.9486	0.9276
BCARNet	0.8757	0.0306	0.8689	0.8650	0.8666	0.9407	0.9365	0.9382
LGIPNet	0.8854	0.0276	0.8791	0.8668	0.8268	0.9499	0.9432	0.9213
Ours	0.8856	0.0249	0.8865	0.8823	0.8845	0.9540	0.9499	0.9537

respectively. We report the quantitative measurements of our method and other comparison methods on the ORSI-4199 dataset in Table II. Overall, our method performs equally well on the ORSI-4199 dataset, with most metrics outperforming all comparison methods. Specifically, the five metrics ($\mathcal{M}$, F_β^{mean}, F_β^{adp}, E_ξ^{mean}, and E_ξ^{adp}) of DiffORSINet are the best, being 0.11%, 0.19%, 0.87%, 0.13%, and 2.61% higher than the second-place method, respectively. The $F_\beta^{\max}$ and $E_\xi^{\max}$ metrics are suboptimal.

Furthermore, in Fig. 5, we present the comparison of the PR curves and F-measurement curves of our method with several of the latest methods. The PR curve of the method we proposed is closest to the upper right corner on all three datasets, and the F-m curve covers the largest area on all three datasets. This indicates that our model has achieved better performance, realizing a higher accuracy rate while maintaining a high recall rate.

2) Qualitative Evaluation: In Fig. 6, we present some visual comparisons between DiffORSINet and several other representative SOTA methods. The experimental results show that the network we proposed can detect significant targets in complex scenes more completely and accurately. Fig. 6 shows five examples, which belong to typical and challenging ORSI scenarios in three datasets, including: 1) shadow targets; 2) multiple small targets; 3) irregularly outlined targets; 4) multiscale targets; and 5) targets in complex backgrounds. Specifically, in the detection of shadow scenes, only our method can detect the complete target. Other methods have problems such as regarding some shadows as targets or incomplete detection of the original target under the interference of shadows. In the detection of multiple small targets, only our method can detect all the small targets completely and clearly. Most of the other methods have the problem of incomplete detection or unclear and nonstandard edges. In the detection of irregular contour targets, our method can capture the complete

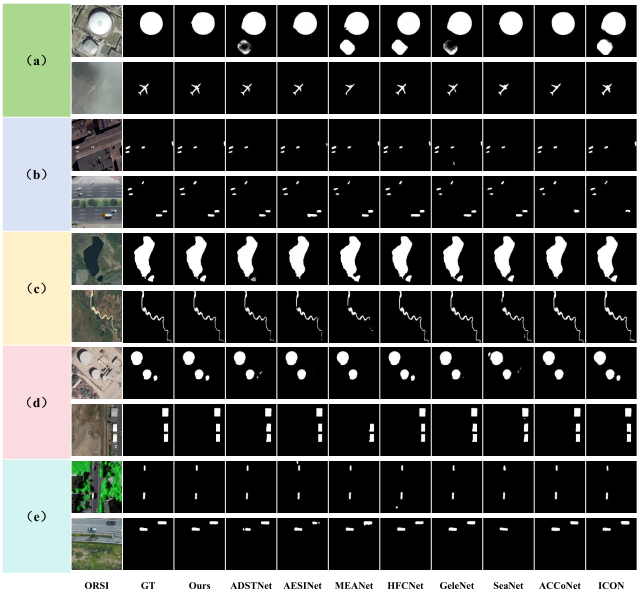


Fig. 6. Visual comparison of our method with several representative significance models. (a) Shadow target. (b) Multiple small targets. (c) Irregularly outlined targets. (d) Multiscale targets. (e) Targets in complex backgrounds.

TABLE III

ABLATION EXPERIMENTS OF DIFFERENT MODULES, INCLUDING FFAM AND MFFM, THE BEST RESULTS IN EACH COLUMN ARE INDICATED IN RED

No.	Baseline	FFAM	MFFM	EORSSD			
				$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$
1	✓			0.9327	0.0054	0.8784	0.9771
2	✓	✓		0.9440	0.0047	0.8978	0.9827
3	✓		✓	0.9448	0.0042	0.8987	0.9814
4	✓	✓	✓	0.9450	0.0038	0.8994	0.9848

shape of the target and achieve detail differentiation. In the detection of complex background objects, our method still has excellent performance, while other methods are prone to false detection problems due to the interference of background noise. We observe that for all these scenarios, DiffORSINet clearly highlighted all significant targets, while other comparison methods are disturbed by the scenarios and might fail in individual cases, resulting in missed false detections.

C. Ablation Studies

We conduct a comprehensive ablation study on the EORSSD dataset to evaluate the effectiveness of each module of DiffORSINet and the important components within each module. We carry out a series of ablation experiments by deleting or replacing individual components. For fair comparison, all ablation studies are retrained using the same parameter settings and training strategies as those in Section IV.

1) Effectiveness of Modules in Conditional Guided Denoising Networks: To evaluate the contribution and impact of each component in DiffORSINet, we design various combinations of different modules and conduct a series of ablation studies on the EORSSD dataset. Specifically, there are the following

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TABLE IV
ABLATION EXPERIMENTS FOR EVALUATING THE EFFECTIVENESS OF EACH COMPONENT IN THE PROPOSED MODULE, WITH THE BEST RESULTS IN EACH COLUMN INDICATED IN RED

No.	Models	EORSSD			
		$S_\alpha \uparrow$	$\mathcal{M} \downarrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$
2	FFAM w/o FFT-2D	0.9444	0.0041	0.8992	0.9830
3	FFAM w/o CA	0.9450	0.0044	0.8990	0.9836
4	FFAM w/o FAU	0.9440	0.0042	0.8991	0.9822
5	MFFM w/o Part 1	0.9446	0.0044	0.8970	0.9812
6	MFFM w/o Part 2	0.9422	0.0047	0.8974	0.9843
1	Net(Ours)	0.9450	0.0038	0.8994	0.9848

variants: 1) baseline, composed of the diffused backbone and ordinary convolutional layers; 2) baseline + FFAM; 3) baseline + MFFM; and 4) baseline + FFAM + MFFM. As shown in the quantitative performance in Table III, we observe that each module of DiffORSINet made a positive contribution to the performance improvement of the baseline network.

Taking the EORSSD dataset as an example, the complete DiffORSINet improves the original baseline by 1.23%, 0.16%, 2.1%, and 0.77%, respectively, on S_α, $\mathcal{M}$, $F_\beta^{\max}$, and $E_\xi^{\max}$. FFAM is an important component of DiffORSINet. After adding FFAM to the baseline, the S_α, $\mathcal{M}$, $F_\beta^{\max}$, and $E_\xi^{\max}$ indicators increase by 1.13%, 0.07%, 1.94%, and 0.56%, respectively. This is because FFAM can further utilize noise mask features to guide the decoding process to perform local modeling and refine the local details of features. After adding MFFM to the baseline, the indicators of S_α, $\mathcal{M}$, $F_\beta^{\max}$, and $E_\xi^{\max}$ increase by 1.21%, 0.12%, 2.03%, and 0.43%, respectively. Low-level features often contain rich detailed information, while high-level features contain location information. Therefore, when conducting the interaction and fusion of high-level and low-level features, we try to capture these details and location information from a global perspective as much as possible. MFFM can mine the intrinsic consistency information of features, model the context, enhance the dominant information in the features, and suppress background noise, thereby bringing performance improvement at the baseline. The above analysis verifies the validity of each module in DiffORSINet, and their combination significantly improves the model performance.

2) Effectiveness of Each Component in FFAM: FFAM contains the important component FAU, which mainly includes the 2-D FFT (FFT-2D) and the CA mechanism. To verify the importance of this component, we design three variants: 1) removing the FAU (i.e., w/o FAU); 2) removing the FFT-2D part (i.e., w/o FFT-2D); and 3) removing the CA mechanism (i.e., w/o CA). As shown in rows 2, 3, and 4 of Table IV, the removal of any component reduces the detection accuracy, indicating that this FAU component and the two parts within it are necessary for FFAM.

The advantage of the Fourier transform lies in converting spatial information into the frequency domain, allowing different frequency components to be processed separately. This global transformation makes feature decoupling physically interpretable, with low frequencies corresponding to macroscopic structures and high frequencies to microscopic details. In the ORSI-SOD task, the target may be confused with the background due to high-frequency details being masked by noise or low-frequency information not being prominent enough. Through frequency-domain decomposition, modules can, respectively, enhance or suppress specific frequency components, thereby handling these issues more flexibly. It can be seen from the experiment that modifying and reducing the internal structure of FAU will lead to a more or less decline in model performance. When there is a lack of components for refining regional and boundary information, especially in FFT-2D, the metrics on the EORSSD dataset are all reduced. Similarly, the CA for enhancing regional information utilizes the overall structural information of low frequencies to guide the selection of high-frequency details, which is equivalent to weighting the details based on the overall structure, enhancing the details related to the target, suppressing noise, and fusing the features of the frequency domain to improve the clarity of the target boundary. Therefore, the CA part is also indispensable.

3) Effectiveness of Each Component in MFFM: MFFM consists of Part 1 and Part 2. Therefore, we provide two variants to study the effectiveness of the components mentioned above: 1) removing Part 1 (i.e., w/o Part 1) and 2) removing Part 2 (i.e., w/o Part 2). As shown in rows 5 and 6 of Table IV, removing either component reduces the detection accuracy, which indicates that both parts are necessary for MFFM.

Part 1 separates high-level features from low-level features through a dual-branch structure. The left branch focuses on high-level semantics, while the right branch focuses on low-level details, avoiding gradient competition and feature flooding caused by direct fusion. Convolution operations are performed on the two branches, respectively, to achieve cross-scale feature alignment within the local receptive field, while the residual structure introduces identity mapping in the initial fusion to ensure the lossless transmission of the original feature information. In Part 2, the statistical patterns of background noise are identified through multiscale pooling after adding the two branches, which is equivalent to constructing a spatial prior distribution. The essence of the weight output by Sigmoid is a posterior probability. This probabilistic processing is particularly suitable for scenarios with high background complexity in remote sensing images. When different high-frequency elements coexist, only the high-frequency information that matches the deep semantics will be enhanced. It can be found from the experiment that removing Part 1 has a significant impact on the $\mathcal{M}$ metric and $E_\xi^{\max}$ metric, and removing Part 2 has a significant impact on the $\mathcal{M}$ metric and S_α metric. Retaining the feature fusion at each level or retaining the further processing part of the two branches after fusion will lead to a decline in performance.

4) Influence of Different Time Steps: Fig. 7 provides the gradual evolution of the prediction results at different time steps. In the initial stage, the model generates uncertain coarse masks for regions with unclear boundaries, reflecting its preliminary prediction of significant target areas. As the number of sampling steps increases, the model gradually establishes clear boundaries through its characteristic of layer-by-layer iteration to achieve precise segmentation. However, if the

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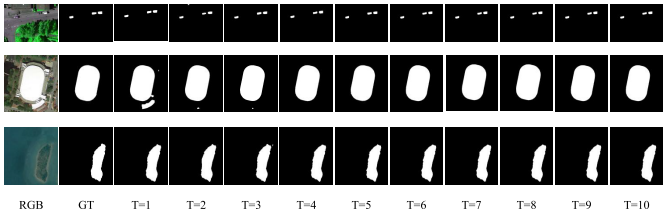


Fig. 7. Visualization results of the model at different time steps.

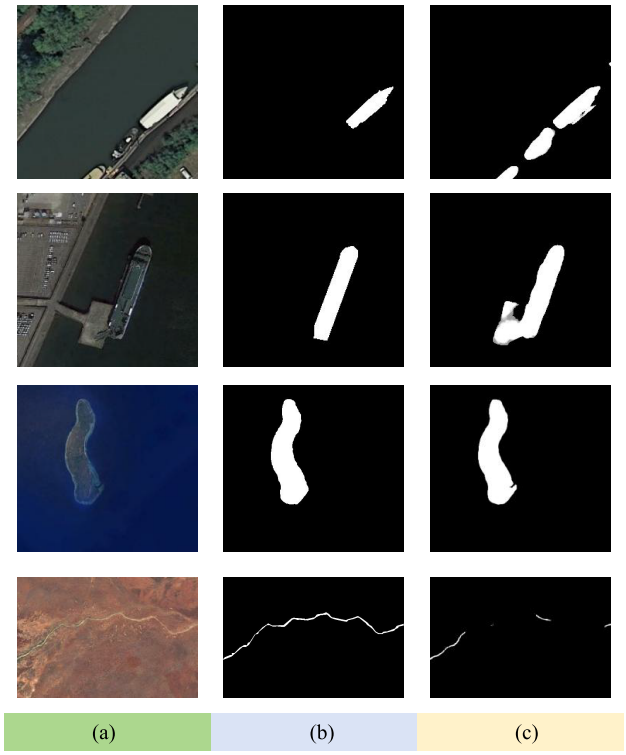


Fig. 8. Some failure cases on the EORSSD dataset. (a) ORSI, (b) GT, and (c) ours.

model has converged and the sampling process becomes stable, different sampling steps may produce similar results. As shown in Fig. 7, most of the sampling results of our model have very small differences. Thus, even with a relatively small number of sampling steps, high-quality results can be produced. Although $T = 10$ can achieve accurate segmentation, the $T = 5$ we set has already achieved good results, which greatly improves the reasoning efficiency.

D. Analysis of Failure Samples

As mentioned earlier, although the DiffORSINet model we proposed has demonstrated outstanding performance in detecting salient object targets in optical remote sensing images, there are still opportunities for further improvement. To emphasize the challenges faced by our model, we present some representative failure cases in Fig. 8. In the first case, as shown in the first two lines of the figure, accurately distinguishing whether buildings constitute significant areas remains a challenging task. When the image contains objects with different semantics, the model sometimes fails to detect the most significant objects. Although our model enhances the interaction between local and global features, preexisting knowledge during the learning process may sometimes lead to incorrect judgments. The second issue is that, as shown in the third and fourth lines of the figure, when the foreground and background are extremely similar, the boundary between the target and its surrounding environment is not clear, which makes it difficult for our model to distinguish the target object from the background. This might be due to the uneven distribution of samples in the training dataset, which causes the model to overfit certain features during the learning process. Moreover, the limitations of the model's semantic understanding ability may also lead to deviations or errors in the results. In the future, these issues still need to be further explored.

V. Conclusion

This article proposes an ORSI-SOD framework DiffORSINet based on the diffusion mechanism. Taking advantage of the conditional generation iterative refinement ability of the diffusion model, this method effectively solves the problems of irregular contours and complex backgrounds of significant targets in ORSI-SOD. The FFAM module in this method makes full use of the Fourier transform operation and the CA mechanism to capture and fuse the frequency-domain features, enhance the signal strength of some parts, thereby refining the image details and achieving effective denoising. Furthermore, by fusing and coordinating multilevel features, MFFM effectively integrates global semantic information and local detail information, suppresses irrelevant regions, and highlights important regions. We conducted a comprehensive experiment on three public ORSI-SOD datasets. The experimental results verify the superiority of the weight network and the effectiveness of each module. Furthermore, future research directions will focus on architecture optimization in complex operating environments to enhance performance. We will also explore lightweight networks that are more suitable for real-world scenarios to optimize the network.

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